

B - Spiral Galaxy

分值: 200 分

Problem Statement

There is a grid with H rows and W columns. The cell at the i -th row from the top and j -th column from the left is denoted as cell (i, j) .

有一个包含 H 行和 W 列的网格。从上往下第 i 行、从左往右第 j 列的单元格记为 (i, j) 。

Each cell of the grid is colored white or black. The information of the grid is given by H strings $S_1, S_2, \dots, S_H$ each of length W : cell (i, j) is white if the j -th character of S_i is `.`, and black if it is `#`.

网格的每个单元格被染为黑色或白色。网格的信息由 H 个长度为 W 的字符串 $S_1, S_2, \dots, S_H$ 给出：若 S_i 的第 j 个字符为 `.`，则单元格 (i, j) 是白色；若为 `#` 则是黑色。

Find the number of rectangular regions of the grid that are point-symmetrically colored.

求网格中满足颜色中心对称的矩形区域的数量。

More formally, find the number of integer tuples (h_1, h_2, w_1, w_2) satisfying all of the following conditions:

更正式地，求满足以下所有条件的整数元组 (h_1, h_2, w_1, w_2) 的数量：

- $1 \leq h_1 \leq h_2 \leq H$
- $1 \leq w_1 \leq w_2 \leq W$
- For all integers i, j satisfying $h_1 \leq i \leq h_2$ and $w_1 \leq j \leq w_2$, cell (i, j) and cell $(h_1 + h_2 - i, w_1 + w_2 - j)$ have the same color.
- $1 \leq h_1 \leq h_2 \leq H$
- $1 \leq w_1 \leq w_2 \leq W$
- 对于所有满足 $h_1 \leq i \leq h_2$ 和 $w_1 \leq j \leq w_2$ 的整数 i, j ，单元格 (i, j) 和单元格 $(h_1 + h_2 - i, w_1 + w_2 - j)$ 颜色相同。

Constraints

- $1 \leq H, W \leq 10$
- H and W are integers.
- S_i is a string of length W consisting of `.` and `#`.
- $1 \leq H, W \leq 10$
- H 和 W 是整数。

- S_i 是一个长度为 W 、由 $.$ 和 $\#$ 组成的字符串。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

```
H W
S1
S2
⋮
SH
```

Output

Output the answer.

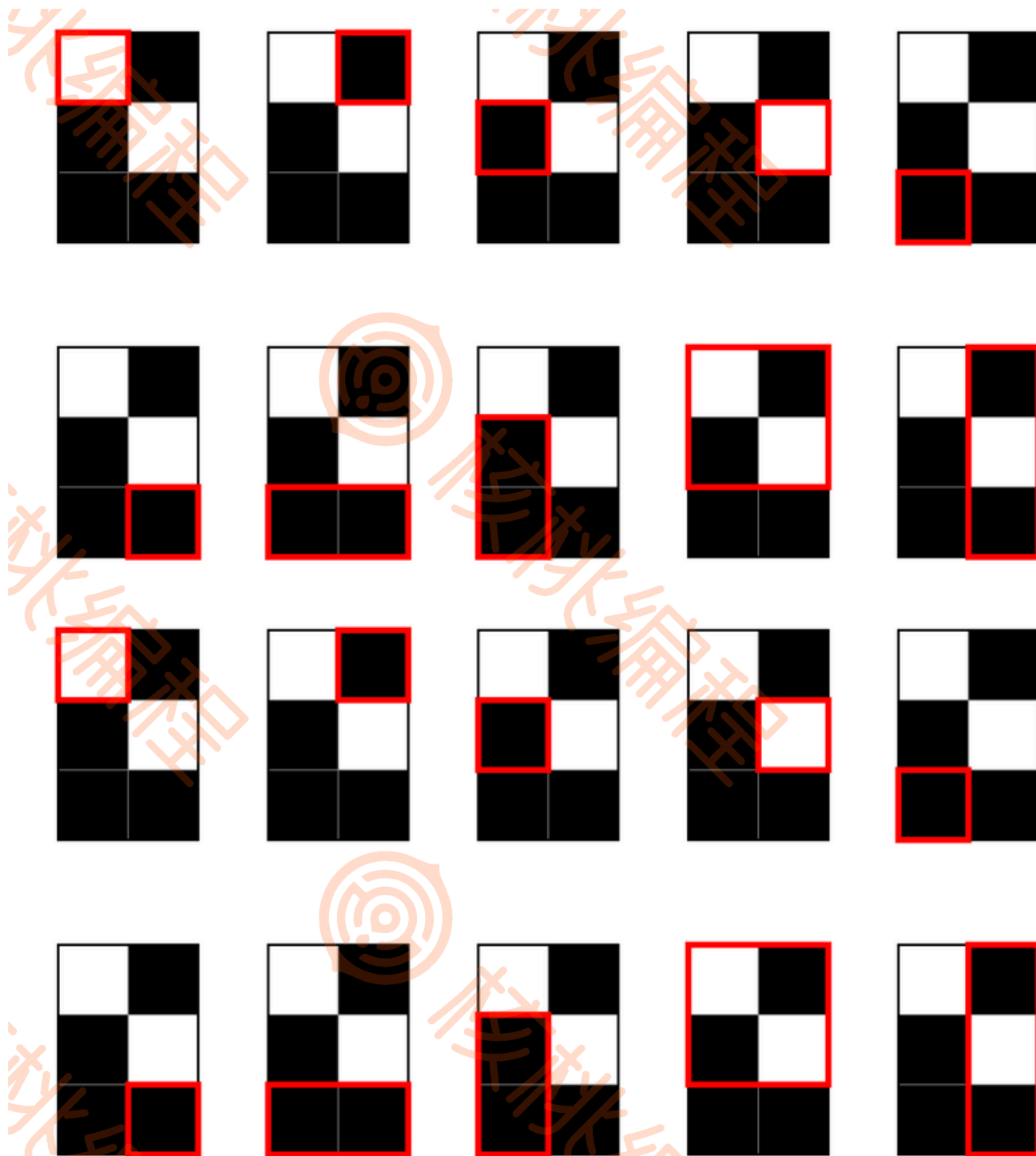
输出答案。

Sample Input 1

```
3 2
.#
#.
##
```

Sample Output 1

```
10
```



As shown in the figure above, the answer is 10.

如上图所示，答案为 10。

Sample Input 2

```
4 5
.#.#.
####.
```

```
##...#  
....#
```

Sample Output 2

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